

PATENTING THE SMART TAPPER IN INDIA

Every problem you can face at the Indian Patent Office, and how it is solved

*Companion to: Form2_Provisional_Specification.docx, Form2_Complete_Specification_DRAFT.docx,
Patent_Drawing_Sheets.pdf*

CONFIDENTIAL. For the inventors and their guide only.

This is practical guidance prepared from the Patents Act, 1970 and the Patents Rules, 2003 as amended (including the 2024 amendment). It is not formal legal advice. Before filing the complete specification, get the documents checked by a registered patent agent, your college IPR cell or a Patent Information Centre. Always check current fees and forms on ipindia.gov.in.

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1. All Problems at a Glance

Status key: FIXED = already solved in the new files. YOU = an action only your team can do.

#	Problem	Law	Solution	Status
1	"Method of tapping a tree" claim is a method of agriculture	Sec. 3(h)	Method claim removed; only device claims	FIXED
2	Software logic (tag memory, interpolation) may be called a computer program per se	Sec. 3(k)	Every claim tied to hardware with a physical technical effect; reply prepared	FIXED
3	Straps, rail, load cell, tag reader are known parts: "mere arrangement"	Sec. 3(f)	Claim 1 recites functional link: probe force → blade position; reply prepared	FIXED
4	Examiner may say obvious: motorised knife + manual bark gauge	Sec. 2(1)(ja)	Arguments + fallback claims + test evidence	FIXED / YOU (tests)
5	Idea shown publicly before filing destroys novelty	Sec. 29-31	File provisional first; confidentiality rules; NDA	YOU
6	Vague words ("about", "preferably", "for example") in claims	Sec. 10(4)(c)	Removed from claims	FIXED
7	Claim 1 broader than description ("gauging means")	Sec. 10(4), 10(5)	Claim 1 now recites the fully described needle + force sensor	FIXED
8	Threshold values unknown: insufficient "best method"	Sec. 10(4)(b)	Calibration paragraphs with blanks to fill from tests	YOU (tests)
9	Trade names (ESP32, Arduino, NTAG213, GT2, NEMA) in specification	Practice / clarity	Generic terms used in Form 2	FIXED
10	Title 19 words and contained "method"	Rule 13(7)(a): max 15 words	New title, 13 words	FIXED
11	Drawings had titles and descriptive text; no sheet details	Rule 15 / drawing format	New A4 drawing sheets with margins, sheet numbers, signature line	FIXED / YOU (name, sign)
12	Specification page format	Second Schedule format	A4, 4 cm top/left, 3 cm bottom/right, 1.5 spacing, numbered paragraphs, claims and abstract on new pages	FIXED
13	12 claims → extra claim fees	First Schedule	Reduced to 10 claims	FIXED
14	Wrong deadline in v2: examination request is now 31 months, not 48	Rule 24B (2024)	Corrected; file Form 18 with the complete specification	FIXED
15	Missing the 12-month complete-specification deadline = application abandoned	Sec. 9(1)	Calendar plan; file complete by month 10	YOU
16	Improvements made after the provisional are not covered	Sec. 9(2)	File a second (cognate) provisional before disclosing improvements	YOU (if needed)
17	No digital signature (DSC) for e-filing	e-filing rules	Get a Class 3 DSC, file through the IPR cell or agent, or file on paper at the Chennai office	YOU
18	Disputes over ownership: students / guide / college	Sec. 6, 7(2)	Sign the inventorship and IP agreement now (Section 5)	YOU
19	Wrong or missing inventors	Sec. 6; Form 5	Only real human contributors; keep a logbook; AI tools cannot be inventors	YOU

#	Problem	Law	Solution	Status
20	Applicant is not the inventor → "proof of right" needed	Sec. 7(2)	Inventors endorse Form 1 or sign an assignment within 6 months	YOU
21	Cost of filing and agent	First Schedule	Low-fee categories, IPR cell / PIC help, SIPP for startups, 10 claims only	FIXED / YOU
22	Slow examination (years)	Rule 24C	Form 18A if eligible (startup, small entity, female applicant, government institution)	YOU (check eligibility)
23	First Examination Report (FER) objections	Sec. 12-15	Reply templates in Section 4; 6 + 3 months to reply	FIXED (templates)
24	Pre-grant or post-grant opposition by a competitor	Sec. 25	Solid claims, test evidence, search record	YOU
25	Filing abroad without permission	Sec. 39	File in India first; wait 6 weeks or get permission; PCT within 12 months	YOU (if exporting)
26	After grant: renewal fees and working statement forgotten	Sec. 53, 146; Form 27	Renewal every year from year 3; Form 27 once every 3 financial years	YOU
27	Selling may infringe someone else's patent	Sec. 48	Freedom-to-operate check before launch	YOU

2. Patentability Problems (What Can Be Patented)

2.1 Section 3(h): method of agriculture

Problem: Section 3(h) says a method of agriculture or horticulture is not an invention. Version 2 had claim 12, "A method of tapping a tree...". Tapping rubber is a plantation practice, so an examiner would very likely reject that claim.

Solution (done): the method claim is removed. All 10 claims are for the **device** (a machine), which is patentable. The operating steps are still described in the description, so nobody can copy the machine and use it.

2.2 Section 3(k): computer program per se / algorithm

Problem: reading a tag, comparing dates and interpolating depth are software steps. Software alone, or a mathematical method, is not patentable in India.

Solution (done): every claim is a physical device. The control unit is claimed only **together with** the probe, force sensor, actuators and blade, and the result is physical: the blade stops short of the cambium, and the bark is probed less often. Under the Indian guidelines for computer-related inventions, a claim is not excluded when software gives a technical effect through hardware. A reply is ready in Section 4.2.

2.3 Section 3(f): mere arrangement of known devices

Problem: straps, rails, load cells, stepper motors and tag readers are all known. Section 3(f) excludes a mere arrangement of known devices each working independently in a known way.

Solution (done): in claim 1 the parts are **not independent**. The probe's force signal is processed to find the bark thickness, and that value **directly sets the blade position**. In claim 3 the tag value replaces probing. This interdependence gives a new result: an automatic cut depth that differs for each tree. See the reply in Section 4.3.

2.4 Inventive step (non-obviousness)

Problem: an examiner may cite a motorised tapping knife (D1) and a manual bark gauge (D2) and say it is obvious to combine them.

Solution:

- **Arguments** (Section 4.4): neither document teaches using a probe's force profile to position a blade automatically. The combination creates a new problem (cambium puncture from probing), which the invention solves by placing the probe inside the band of bark removed by the cut and by storing values on the tree so probing is infrequent. That is not obvious.
- **Fallback claims:** if claim 1 is rejected, combine claim 1 with claim 3 (tag memory) or claim 2 (probe in the removed band) or claim 5 (alignment pins). These narrower combinations are much harder to call obvious.
- **Evidence (YOU):** test data showing better depth accuracy, fewer wounds and equal or better yield proves a technical advance. Fill in the calibration and field-result paragraphs in the complete specification. Evidence often decides inventive-step disputes.
- **Search (YOU):** do the prior-art search in the project file before filing. If you find something close, adjust claim 1 first rather than let the examiner find it.

2.5 Loss of novelty by your own disclosure

Problem: anything made public before your filing date, including by you, counts against novelty. Examples: a project expo, a paper or conference, a YouTube or Instagram post, a GitHub upload, a published project report, a demonstration in a public field, a competition entry, or sharing full drawings with vendors without confidentiality.

Solution (YOU):

1. File the provisional application **before** any public showing.
2. Mark all documents "CONFIDENTIAL"; keep the project report out of the library or repository until filing.
3. Give vendors or fabricators only individual part drawings, and ask them to sign a simple confidentiality letter (Section 6).
4. Field tests: test on private land with the owner's written agreement to keep it confidential.
5. Section 31 allows only limited exceptions (for example certain government-notified exhibitions or papers read before a learned society, then filing within 12 months). Do not rely on it. File first.

3. Drafting and Format Problems (All Fixed)

Item	Rule / requirement	What the new files do
Title	Short, specific, no more than 15 words, no trade marks or fancy words	"A portable rubber tree tapping device with bark-thickness sensing and automatic cut-depth control" (13 words)
Preamble	Provisional: "The following specification describes the invention." Complete: "...particularly describes the invention and the manner in which it is to be performed."	Correct wording in each file
Page layout	A4; margins 4 cm top and left, 3 cm bottom and right; 1.5 line spacing; pages numbered	Applied in both Form 2 files
Paragraphs	Numbered for easy reference in FER replies	[001], [002] ...
Clarity of claims	No "about", "preferably", "for example", "such as", "substantially" in claims	Removed from claims; kept only in the description
Support	Claims must be fairly based on the description	Claim 1 recites the needle probe and force sensor fully described in [025]-[026]
Sufficiency / best method	Complete specification must enable a skilled person and disclose the best method	Calibration and field-result paragraphs ready for your data
Trade names	Avoid brand/product names in the specification	Generic terms ("microcontroller", "near-field communication tag", "toothed belt")
Unity	One inventive concept	Claims 2 to 10 all depend on claim 1
Claims count	Fee payable for each claim above 10	Exactly 10 claims
Claims page	Start on a new page with "We claim:"; signature and date after claims	Done
Abstract	No more than 150 words, on a separate page, indicate the figure	137 words, ends with "Figure 2"
Drawings	A4, same margins, black lines, reference numerals, no descriptive matter except where needed (flow chart, block diagram); applicant name, total sheets, sheet number, signature on each sheet	Patent_Drawing_Sheets.pdf (5 sheets). You write the name and sign.
Reference numerals	Same numbers in drawings and text	Checked: 10 to 108 consistent

4. Ready-Made Replies to Likely Examiner Objections

Adapt these when you receive the First Examination Report (FER). The reply is due within **6 months** of the FER, extendable by **3 months** with Form 4. Always attach amended claims (clean and marked-up copies) and request a hearing "in case the objections are not waived".

4.1 Objection: lack of novelty or inventive step (D1 = motorised knife, D2 = bark gauge)

The Applicant respectfully submits that none of the cited documents, alone or in combination, discloses or suggests a control unit that advances a needle probe through a force sensor, determines the bark thickness from the probe travel between first contact and the rise in force at the wood, and drives a depth actuator to position a blade at that thickness less a residual thickness (claim 1).

D1 discloses a fixed mechanical depth guide; the depth does not depend on the bark of the individual tree. D2 is a hand-held gauge used for inspection and is not connected to any cutting means. A person skilled in the art combining D1 and D2 would at most obtain a knife and a separate gauge used manually; there is no teaching to process the probe force signal to position the blade automatically.

Further, automatic probing would itself cause repeated punctures of the cambium. The invention overcomes this by locating the probe within the band of bark removed by the cut (claim 2) and by storing the sensed thickness on a data carrier on the tree so that probing is needed only at intervals (claim 3). Neither problem nor solution is suggested by D1 or D2.

The technical advance is shown by the test results in paragraphs [0xx]-[0xx]: residual bark $\pm$ mm against $\pm$ mm manually, and fewer cambium wounds per 100 trees. The claimed invention therefore involves an inventive step under Section 2(1)(ja).

4.2 Objection: Section 3(k) (computer program per se)

The claims are directed to a tapping device comprising a guide rail, a motor-driven carriage, a blade, a depth actuator, a needle probe and a force sensor. The control unit is claimed only in combination with these physical elements. It acts on the force signal from the physical probe and controls the physical position of the blade, producing the technical effect that the blade stops short of the cambium of the particular tree, and (claim 3) that the bark is punctured less often. The invention is not a computer program per se, and Section 3(k) does not apply.

4.3 Objection: Section 3(f) (mere arrangement)

In the claimed device the elements do not function independently. The output of the needle probe and force sensor is processed to determine the bark thickness, and that value determines the position of the blade through the depth actuator. Neither the probe nor the blade alone, nor both working separately, would achieve an automatically varying depth of cut adapted to each tree. The combination produces a working interrelationship and a new result, and is not a mere arrangement of known devices under Section 3(f).

4.4 Objection: claims unclear or not supported (Section 10(4)/(5))

The Applicant has amended claim __ to recite __, which is supported by paragraph [0__] and Figure __. The expressions objected to have been deleted. The amended claims are clear, succinct and fairly based on the matter disclosed in the specification.

4.5 Objection: Section 3(h) (method of agriculture)

The claims on record are directed exclusively to a tapping device (an apparatus). No method of agriculture or horticulture is claimed. Section 3(h) is therefore not attracted.

4.6 Other routine requirements in an FER

- Form 3 update (details of any foreign filings) within the time the FER specifies.
- Proof of right, if the applicant is not the inventor.
- Correcting drawings or the abstract: easy to do. Submit replacement sheets.

5. Ownership and Inventorship (Avoiding Disputes)

Problems: (a) the idea came from your teacher, the design work from students, and the college may have an IP policy; (b) later someone leaves or disagrees; (c) the applicant (for example the college or your company) is not the inventor, so proof of right is needed.

Rules:

- **Inventors** are only the natural persons who contributed to the inventive features in the claims. AI tools cannot be inventors, and help from any tool, book or website does not make it an inventor. Every person you list must understand and be able to explain the invention. Your team should do the real development, testing and calibration and record it in a dated logbook.
- **Applicant** can be the inventors, the college, a company, or a mix.
- If the applicant is not the inventor, **proof of right** is needed: the inventors sign the endorsement in Form 1 or an assignment deed, filed within 6 months (Section 7(2)).
- Changing the applicant later (for example to your startup): before grant with **Form 6**; after grant, register the assignment with **Form 16**.

5.1 Template: Inventorship and IP Agreement (sign before filing)

INVENTORSHIP AND INTELLECTUAL PROPERTY AGREEMENT

Date: _____ Place: _____

Parties: (1) _____ (2) _____ (3) _____
(4) _____ (students) and (5) _____ (guide), of _____ College.

1. Invention: "A portable rubber tree tapping device with bark-thickness sensing and automatic cut-depth control" as described in the draft dated ____.
2. Inventors: The parties agree that the inventors are: _____.
3. Applicant: The patent application shall be filed in the name(s) of _____ [the inventors / the College / both].
4. Costs: Fees shall be paid by _____ in the ratio _____.
5. Benefits: Income from licensing or sale of the patent shall be shared _____ [as per the College IPR policy / equally / in the ratio ____].
6. Confidentiality: No party shall disclose the invention publicly before the provisional application is filed, and thereafter only as agreed.
7. Future assignment: The parties agree to assign their rights to a company formed to commercialise the invention on terms to be agreed in writing, with the College's consent where its IPR policy requires it.
8. Leaving: A party who leaves the project remains an inventor for the contribution made, and shall sign any document needed for the application.

Signatures: _____

Witness: _____

6. Confidentiality Letter for Fabricators and Vendors

CONFIDENTIALITY UNDERTAKING

I/We, _____ (fabricator/vendor), have received drawings and information from _____ for fabrication of parts. I/We shall keep them confidential, use them only to make the ordered parts, not copy or disclose them to anyone, and return or destroy them on completion. All rights in the design remain with _____.

Signature: _____ Name / Firm: _____ Date: _____

7. Procedure Problems and the Correct Filing Plan

7.1 Where and how to file

- **Appropriate office:** determined by the applicant's residence or place of business (or the first-named applicant). For Kerala, Tamil Nadu, Karnataka, Andhra Pradesh, Telangana, Puducherry and Lakshadweep it is the **Patent Office, Chennai**.
- **E-filing** (cheaper, faster) needs a **Class 3 digital signature certificate** (DSC) for the person signing. Options: buy one (about ₹1,000 to ₹2,000), or file through your college IPR cell or an agent, or file **on paper** at the Chennai office (paper fees are about 10 % higher).
- You may file **without an agent** as an applicant in person. An agent is optional (and needs Form 26).

7.2 Documents to upload with the provisional application

Document	Notes
Form 1: application	Names of all applicants and inventors; signed by all applicants; inventors' endorsement if applicant ≠ inventor
Form 2: provisional specification	Form2_Provisional_Specification (fill names, sign, save as PDF)
Drawings	Patent_Drawing_Sheets.pdf (fill name, sign each sheet)
Form 3: foreign filing statement	State "no foreign application"; update if you file abroad later
Form 5: declaration of inventorship	Can be filed with the complete specification (or within 1 month after)
Form 28	Only if claiming startup or small-entity status (with proof)
Proof of eligibility for lower fees	For example educational institution documents, if the college is the applicant
Fee	Paid online at filing

7.3 Timeline with deadlines (Day 0 = provisional filing date)

When	Action	Form
Before Day 0	Search, agreement signed, documents finalised, DSC ready	-
Day 0	File provisional + drawings	Forms 1, 2, 3 (+28)
Day 0 to month 9	Build, test, calibrate; keep improvements confidential	-
Any time before disclosing a major improvement	Second (cognate) provisional if needed	Forms 1, 2
Month 10 (latest month 12, strict)	File complete specification with results + request examination	Forms 2, 5, 18 or 18A
Month 18	Automatic publication (or earlier with Form 9)	(Form 9)
Examination request deadline	31 months from priority date (2024 rules). Filing it at month 10 avoids risk	Form 18
After FER	Reply within 6 months (+3 months with Form 4)	Reply, Form 4
Grant, then from year 3	Renewal fees every year (can be paid in advance)	Renewal
After grant	Statement of working once every 3 financial years	Form 27

7.4 Speeding up (Form 18A)

Expedited examination (Rule 24C) is available to certain applicants. Examples: a DPIIT-recognised startup, a small entity, a female applicant who is a natural person (for joint applicants, check that the rule's conditions are met for all), a government department, or an institution established by a Central or

State Act or funded or controlled by the government (**a government engineering college or state university may qualify**). Check the current Rule 24C text before choosing Form 18A.

7.5 Cost plan (approximate official fees, e-filing, natural person / startup / small entity / educational institution)

Stage	Fee (₹)
Form 1 (provisional)	1,600
Complete specification (≤ 30 pages, ≤ 10 claims)	No extra
Form 18 (or Form 18A)	4,000 (or 8,000)
Form 9 early publication (optional)	2,500
Class 3 DSC	about 1,000 to 2,000
Total to grant without agent	about 6,000 to 12,000
Renewal (years 3 to 6)	about 800 per year, rising later

Lower your cost: use the college IPR cell or a state Patent Information Centre (for Kerala, the KSCSTE Patent Information Centre) for free or low-cost help; DPIIT-recognised startups can get agent fees supported under the SIPP scheme; keep to 10 claims and 30 pages.

8. Problems After Filing and After Grant

- **Opposition:** after publication anyone can file a pre-grant representation (Section 25(1)); after grant, a post-grant opposition within 1 year (Section 25(2)). Defence: accurate claims, a documented search, test evidence.
- **Foreign filing:** Indian residents must not file abroad first. File in India, then wait 6 weeks with no secrecy direction, or get permission under Section 39. File a PCT application within 12 months of Day 0 to keep priority in rubber-producing countries.
- **Renewal:** missing a renewal fee makes the patent lapse (a 6-month grace with extra fee applies; restoration is possible but difficult).
- **Working (Form 27):** file once every 3 financial years. If the invention is not worked in India for 3 years after grant, others may seek a compulsory licence (Section 84). Plan to manufacture or license.
- **Enforcement:** a patent is enforced by a civil suit, which is costly. Keep the evidence file (dated logbook, filings, test data). Licensing to an established manufacturer is often the practical route for students.
- **Freedom to operate:** your patent does not stop others' patents applying to your product. Get an FTO check before selling.

9. Final Pre-Filing Checklist

1. Prior-art search done and recorded; claim 1 checked against the closest result.
2. Inventorship and IP agreement signed by all team members and the guide; college IPR policy checked.
3. Applicant(s) decided; proof-of-right plan ready if applicant $\neq$ inventor.
4. Form2_Provisional_Specification: names, addresses, date and signature filled; saved as PDF.
5. Patent_Drawing_Sheets.pdf: applicant name written, each sheet signed.
6. Form 1 and Form 3 filled; Form 28 if claiming startup / small-entity status.
7. DSC ready (or IPR cell / agent / paper filing arranged).
8. Nothing about the machine has been made public; confidentiality letters with vendors.
9. Calendar: Day 0 + 10 months = file complete specification + Form 18/18A.
10. Test plan started so that the calibration and field-result blanks can be filled.

New files in the folder: Form2_Provisional_Specification.docx (file now),
Form2_Complete_Specification_DRAFT.docx (file within 12 months, after filling test data),
Patent_Drawing_Sheets.pdf, and this guide.